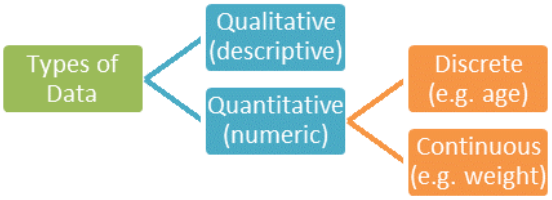
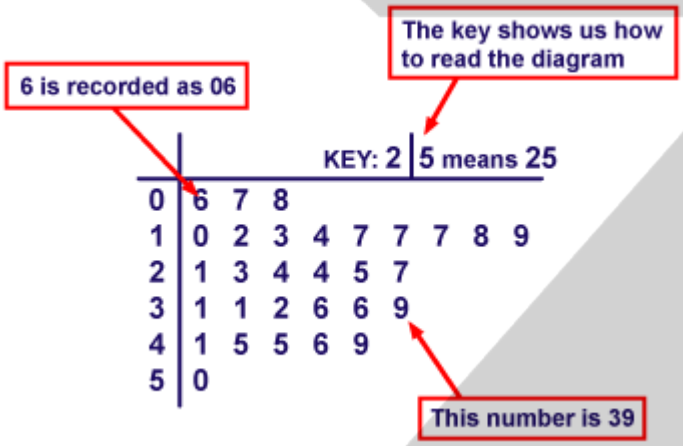
1. Representation of Data

1.1. Types of Data



1.2. Stem-and-Leaf Diagrams

- Used to represent data in its original form
- Each piece of data split into 2 parts; stem & leaf
- Leaf can only by 1 digit and should be written in ascending order
- Always include a key on your diagram



- **Advantage:** contains accuracy of original data

Back-to-back stem-and-leaf diagrams

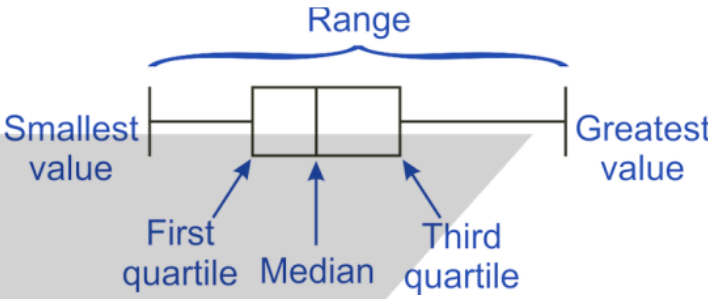
- Note the order of the left side

Female		Male
7 6	0	1 4 5
9 8 8 4 2	1	0 2 3 4 4 8
6 6 5 5 5 4	2	2 7 7 9
5 3 3 3 2 1 0	3	0 0 0 3 6 8
	4	0

Key : 3 | 1 | 4 represents
13 Female
14 Male

1.3. Box-and-Whisker Plots

- Five figure summary:
  - Lowest and highest values
  - Lower and upper quartiles
  - Median
- Mean & standard deviation most useful when data roughly symmetrical & contains no outliers
- Median and interquartile range typically used if data skewed or if there are outliers.



- **Advantage:** easily interpreted and comparisons can easily be made.

1.4. Histograms

- A bar chart which represents continuous data
- Bars have no space between them
- Area of each bar is proportional to frequency

$$Frequency = Frequency\ Density \times Class\ Width$$

- For open ended class width, double the size of previous class width and use this
- If range '$0-9$' then class width is '$-0.5 \le x \le 9.5$'

1.5. Cumulative Frequency Graphs

- Upper quartile $= 75$ th percentile value
- Lower quartile $= 25$ th percentile value
- Median $= 50$ th percentile value

$$Interquartile\ Range = Upper\ Quartile - Lower\ Quartile$$

- When finding the median and quartiles, draw in vertical and horizontal dashed lines on the graph, and find the corresponding x-axis value(s)
- Points are plotted at upper class boundaries
- Join points together with straight lines unless asked to draw a cumulative frequency curve

Key Points:

1. Add up preceding frequencies in the table to get the cumulative frequency
2. Plot points at the upper limit of the class interval

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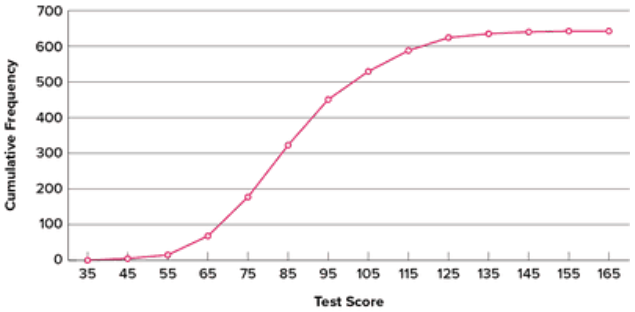
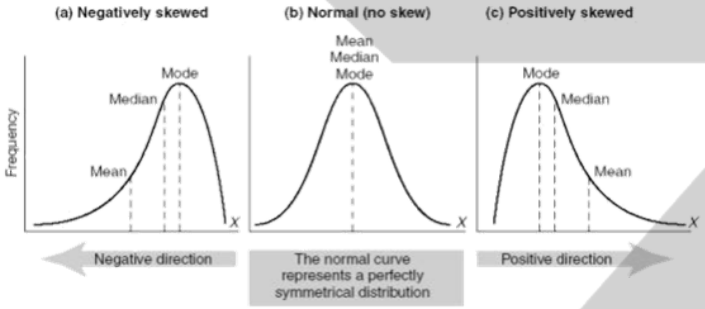
3. Ensure the curve through the points is smooth



1.6. Skewness

- **Symmetrical:** Median line lies in the middle of the box (i.e. UQ – median = median – LQ)
- **Positively skewed:** Median line lies closer to LQ than UQ (i.e. UQ – median > median – LQ)
- **Negatively skewed:** Median line lies closer to UQ than LQ (i.e. UQ – median < median – LQ)



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